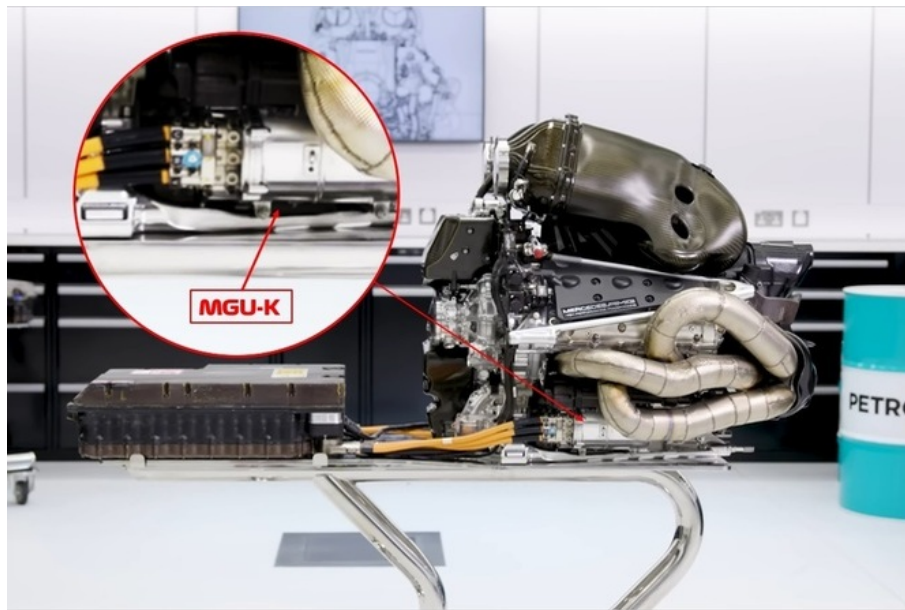

Energy Management in a Formula One Hybrid Power Unit

MGU-H, turbo lag and experimental investigation

Thermodynamic modelling and scale-model validation of
turbocharger-shaft energy recovery



Mercedes Formula One power unit with the MGU-K highlighted. Image: Mercedes-AMG High Performance Powertrains, reproduced via [FormulaPassion](#).

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Scientific report

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Abstract

The turbocharged hybrid power units used in Formula One from 2014 to 2025 combined two electrical machines. The crankshaft-coupled Motor Generator Unit–Kinetic (MGU-K) recovered braking energy and supplied propulsion torque, while the turbocharger-coupled Motor Generator Unit–Heat (MGU-H) either extracted excess turbine power or motored the compressor. The latter function reduced turbo lag: during a throttle transient, exhaust flow and turbine torque rise only after the engine has produced more gas, whereas the compressor must accelerate immediately to establish boost.

This report develops the complete electrical and thermodynamic context of the system, then studies energy extraction on a small-scale turbine–generator bench. A steady-flow model links turbine output, compressor demand, rotor acceleration and electrical power. The original simulation displays a low-speed power deficit followed by a high-speed surplus, which corresponds respectively to MGU-H motor and generator operation. In the experiment, changing the resistive load from 100 Ω to 3 k Ω changes both rotor speed and electrical power. The measured maximum is approximately 0.60 W at 700 Ω ; the original analytical model predicts a broad maximum near 800 Ω . This result is explained through the competition between internal Joule losses, electromagnetic braking and mechanical losses. The study shows that an MGU-H cannot be optimized as an isolated generator: electrical torque must be scheduled with boost demand, rotor dynamics, the MGU-K and the energy store over the complete duty cycle.

Keywords: Formula One, MGU-H, MGU-K, turbo lag, exhaust-energy recovery, electric machine, load matching

Nomenclature

Symbol	Meaning	Symbol	Meaning
P_t, P_c	turbine output, compressor demand	P_H	MGU-H shaft power
P_R	useful load power	P_J, P_f	Joule and mechanical losses
$\dot{m}_a, \dot{m}_g$	air and exhaust mass flow	J, ω	shaft inertia and speed
e, i	machine EMF and current	R, r	load and armature resistance
k_e, k_t	EMF and torque constants	μ	viscous-friction coefficient

1 Introduction

A turbocharger recovers part of the exhaust-gas enthalpy to compress the intake air. It increases the specific power of an internal-combustion engine, but it also creates a transient problem. After the driver requests torque, the compressor and its rotating assembly must accelerate before the desired boost pressure is available. Turbine torque, however, depends on an exhaust flow that is itself produced only after combustion has increased. This causal delay is turbo lag. At high engine speed the opposite situation can occur: the turbine receives more power than the compressor requires, and a conventional wastegate bypasses part of the gas to control boost [1, 2].

The 2014 Formula One regulations placed an electrical machine on the turbocharger shaft. The MGU-H could accelerate the compressor electrically during the power-deficit phase and generate electricity during the surplus phase. It formed part of a larger hybrid architecture with the MGU-K, energy store and power electronics [3, 4]. The scientific question of this project is therefore:

Under which mechanical and electrical conditions is useful energy recovery from a turbocharger shaft maximized without compromising turbocharger response?

The work combines a system description, a control-volume model, a DC-machine model and a turbine-generator demonstrator. The bench does not reproduce Formula One pressure, temperature or speed; it isolates the electromechanical coupling that creates an optimal generator load.

2 Electrical architecture of the Formula One power unit

2.1 Main components

Figure 1 is the power-unit view used in the original presentation. The internal-combustion engine (ICE), turbocharger and two electrical machines are mechanically distinct but energetically coupled. The MGU-K is connected to the crankshaft through a fixed-ratio mechanical transmission. The MGU-H is mounted on the shaft joining the exhaust turbine to the intake compressor. Both machines are connected through bidirectional inverters to a high-voltage DC link and the energy store (ES).

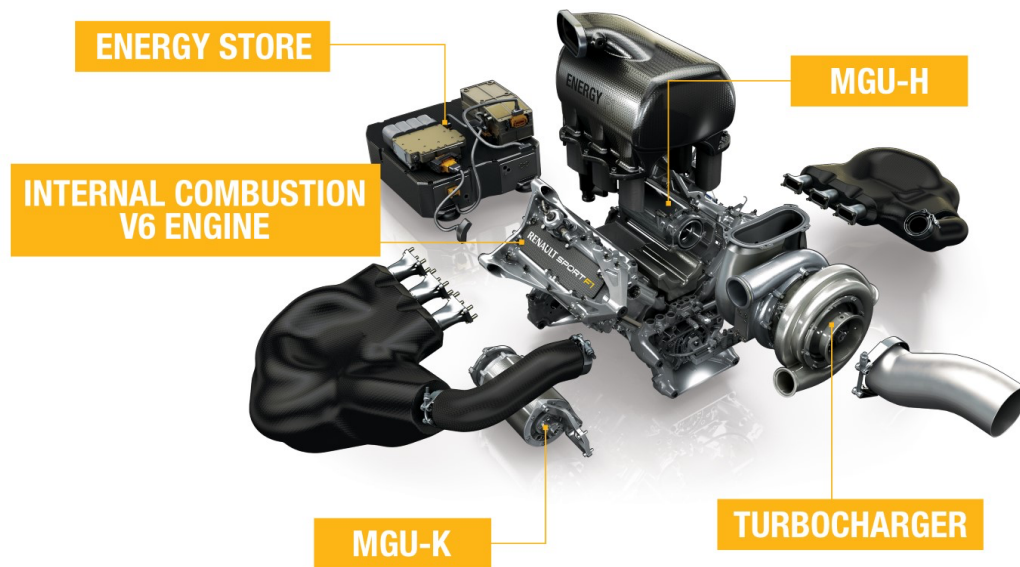


Figure 1. Principal components of the 2014–2025 hybrid power unit.

Source image from the original presentation; Renault Sport F1 illustration [5].

The names can be misleading. The MGU-H does not directly convert heat into electricity; it converts mechanical shaft power that the turbine has obtained from the exhaust-gas enthalpy. Likewise, the MGU-K converts mechanical energy at the crankshaft. In both cases a three-phase electrical machine and an inverter provide controlled bidirectional torque.

2.2 MGU-K: braking recovery and propulsion

During braking, the rear wheels drive the powertrain and the MGU-K operates as a generator. Its electromagnetic torque contributes to deceleration while electrical power flows through the inverter to the energy store. During acceleration, the power flow reverses: the inverter supplies the MGU-K, which adds torque to the crankshaft. In the original 2014 rules its mechanical output was limited to 120 kW, its speed to 50 000 rpm, energy recovered by the MGU-K to 2 MJ per lap, and energy sent from the store to the MGU-K to 4 MJ per lap [3, 4]. These limits made energy routing through the MGU-H strategically important.

2.3 MGU-H: two operating quadrants

The MGU-H has two functions:

- **Motor mode.** Electrical power from the energy store, or directly from the MGU-K through the DC link, produces shaft torque. The compressor reaches its target speed sooner and boost response improves.
- **Generator mode.** When turbine power exceeds compressor demand, electromagnetic torque absorbs part of the surplus. The electricity can charge the store or feed the MGU-K without first being stored.

The direct electrical path is important because every storage cycle adds conversion loss and is constrained by state of charge. Power electronics regulate torque, not a fixed resistance. The

electrical load seen by the turbo shaft can therefore be changed continuously.

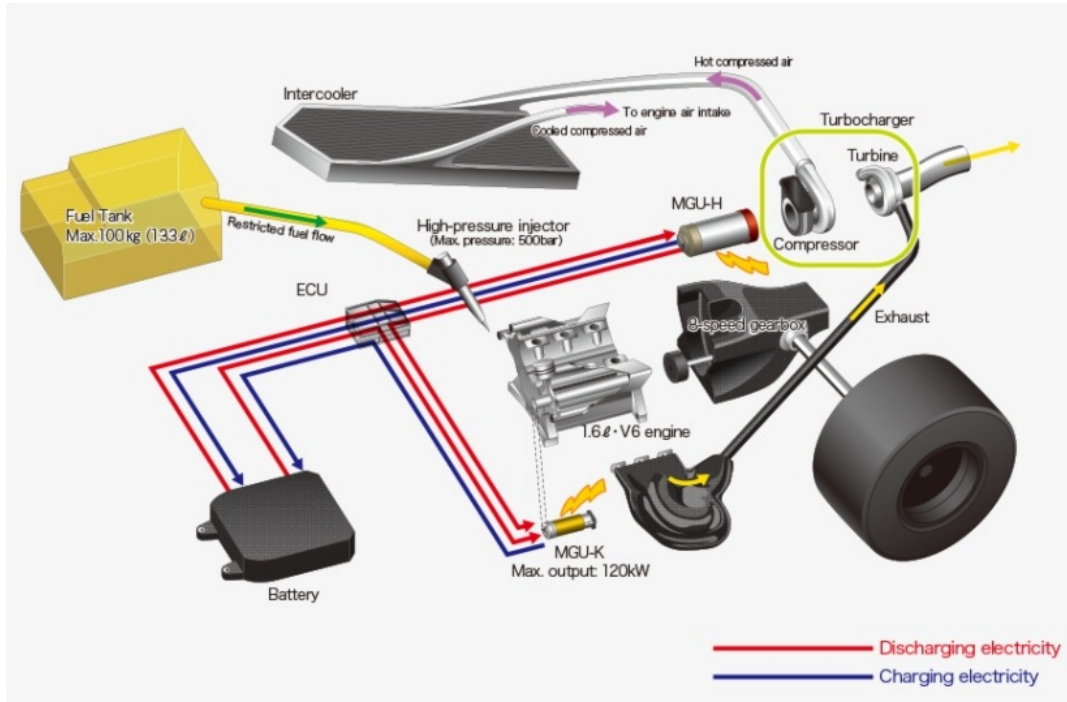


Figure 2. Electrical power paths between the MGU-H, MGU-K and energy store.

Diagram retained from the original presentation. Red paths discharge the store; blue paths recover electrical energy.

Figure 3 reformulates those paths as an energy-flow diagram. It emphasizes that the two motor-generators share one controlled DC bus even though they act on different shafts.

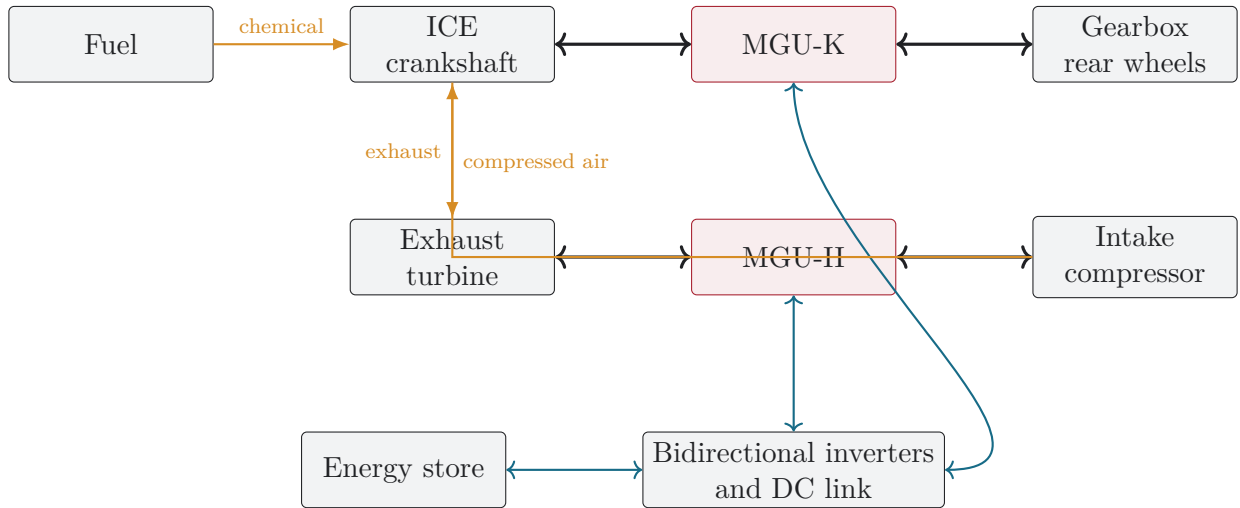


Figure 3. Functional architecture and bidirectional energy conversion.

Four recurring operating states follow from this architecture: kinetic recovery under braking, electrical spool-up during re-acceleration, combined ICE–MGU-K propulsion, and simultaneous MGU-H recovery with MGU-K deployment on a straight. The commands change continuously rather than switching between four ideal states. Figure 4 reproduces the operating-mode illustration used in the oral presentation.

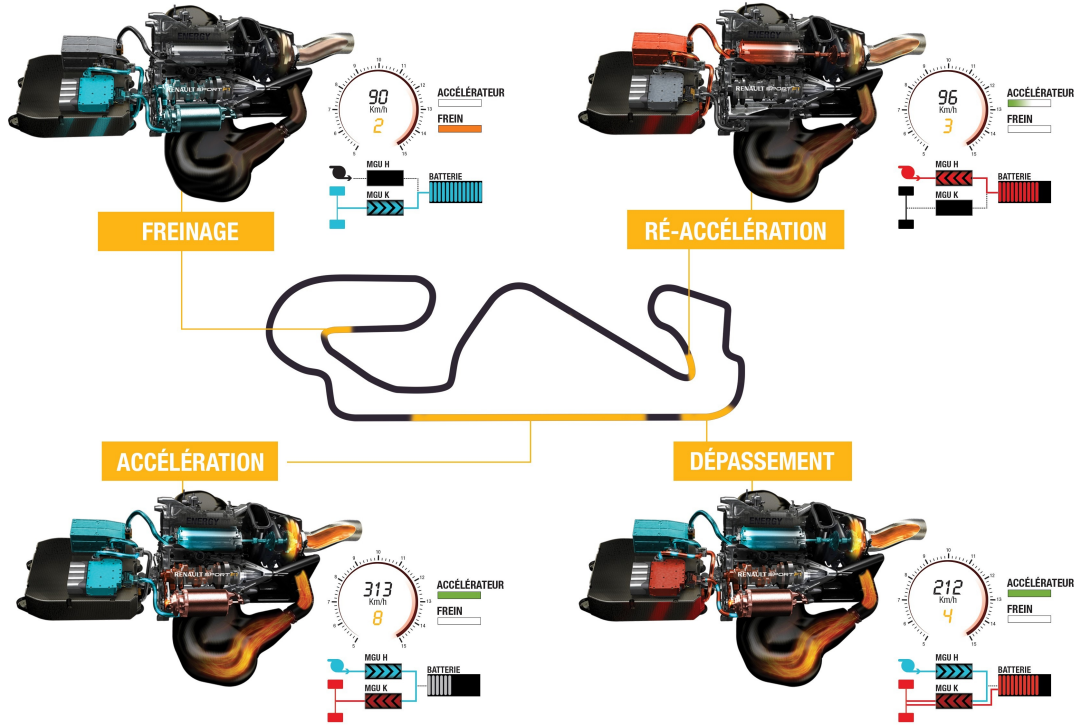


Figure 4. Illustrative hybrid operating modes around a lap.

Original presentation graphic. Blue denotes recovered energy and red denotes deployed energy.

3 Turbo lag and thermodynamic model

3.1 Why turbo lag occurs

The turbocharger rotor has inertia. Its acceleration obeys a torque balance, but the useful physical interpretation is a power balance:

$$J\omega\dot{\omega} = P_t - P_c - P_H - P_{\text{loss}}, \quad (1)$$

where $P_H > 0$ denotes MGU-H generation and $P_H < 0$ motor assistance. Immediately after a throttle request, P_t may remain below the sum of compressor demand and acceleration power. The shaft accelerates slowly, intake pressure rises late and engine torque follows the driver's request with a delay. A larger turbine can recover more high-speed exhaust power but generally increases rotating inertia and low-speed mismatch; a smaller turbine responds earlier but reaches its flow limit sooner [1].

Figure 5 is not a generic replacement curve. It is a translated and annotated reconstruction of the project curve from slide 5 and its archived Python script. The imposed delay in turbine mass flow produces a visible low-speed deficit. The curve is qualitative because it uses fixed pressure ratios and does not include compressor or turbine maps, yet it correctly illustrates the sign change that sets the two MGU-H modes.

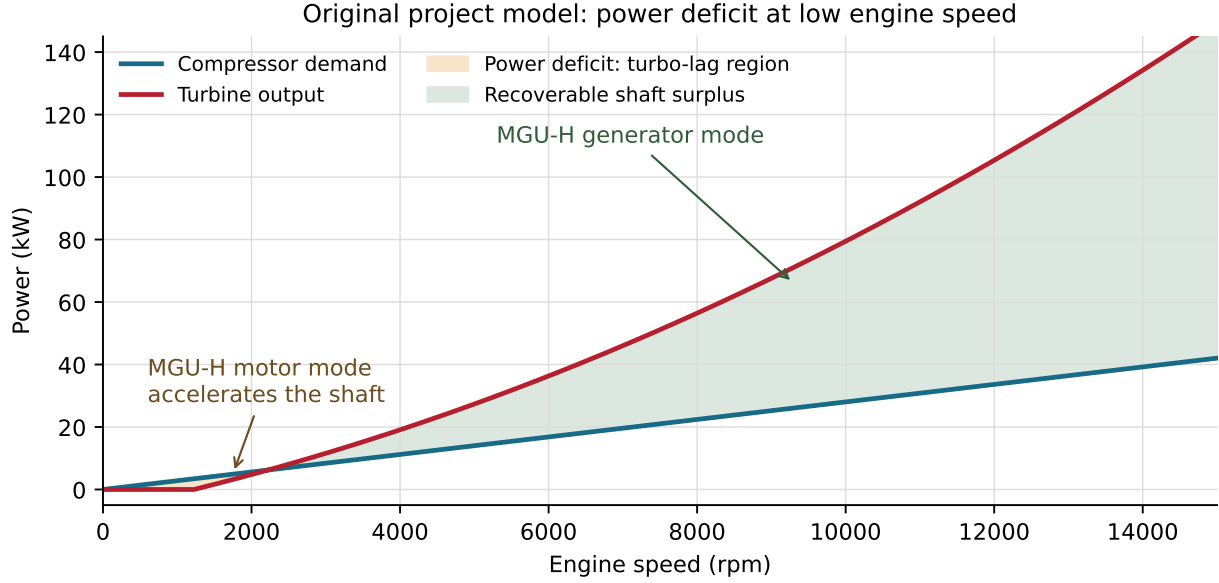


Figure 5. Low-speed power deficit and high-speed shaft surplus in the original model.

In the shaded orange region the turbine alone cannot provide the desired compressor power. A commanded negative P_H adds electrical torque and reduces spool-up time. Once the turbine curve exceeds compressor demand, positive P_H can recover part of the green surplus. Extraction is limited during transients because Equation (1) must still leave enough power for $J\omega\dot{\omega}$.

3.2 Thermodynamic control volumes

Figure 6 reconstructs the thermodynamic scheme from the presentation. The compressor, combustion chamber and turbine are treated as steady-flow control volumes. The compressor and turbine share a shaft with the MGU-H; the combustion chamber changes both mass flow and gas temperature.

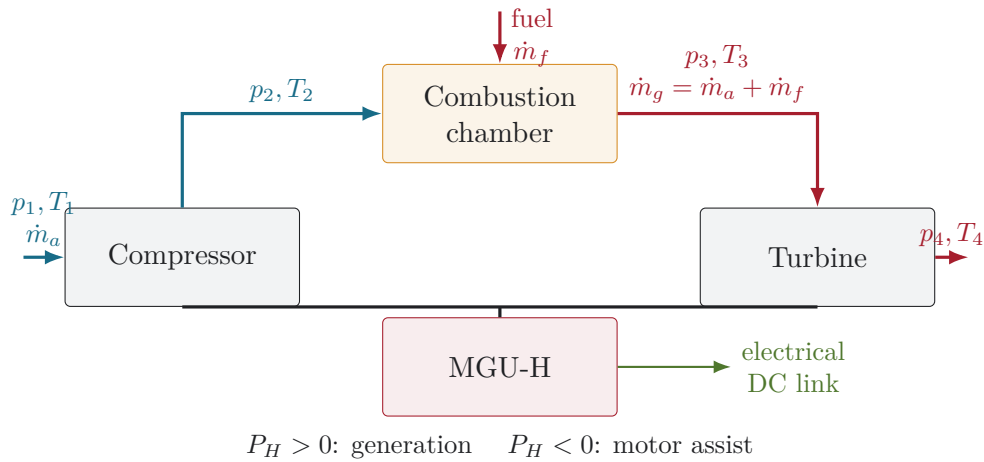


Figure 6. Thermodynamic scheme of the electrically assisted turbocharger.

Neglecting changes in potential energy, inlet-to-outlet kinetic energy and casing heat transfer, the component powers are

$$P_c = \dot{m}_a c_{p,a} (T_2 - T_1), \quad (2)$$

$$P_t = \dot{m}_g c_{p,g} (T_3 - T_4). \quad (3)$$

With pressure ratio $\Pi_c = p_2/p_1$, isentropic efficiencies η_c and η_t , and ideal-gas exponents γ_a and γ_g ,

$$T_{2s} = T_1 \Pi_c^{(\gamma_a-1)/\gamma_a}, \quad T_2 = T_1 + \frac{T_{2s} - T_1}{\eta_c}, \quad (4)$$

$$T_{4s} = T_3 \left(\frac{p_4}{p_3} \right)^{(\gamma_g-1)/\gamma_g}, \quad T_4 = T_3 - \eta_t(T_3 - T_{4s}). \quad (5)$$

For a four-stroke engine, a first estimate of air flow is

$$\dot{m}_a \simeq \eta_v \rho_a V_d \frac{N}{2 \times 60}. \quad (6)$$

This is a mean-value calculation. A predictive model would require corrected flow, compressor and turbine maps, exhaust pulsations, thermal losses and a physically identified rotor inertia [2, 6].

3.3 Recoverable energy

From Equation (1), the instantaneous shaft power available for generation is

$$P_{H,\text{avail}} = P_t - P_c - J\omega\dot{\omega} - P_{\text{loss}}. \quad (7)$$

The positive part of this balance gives a first estimate of recovered electrical energy over a duty cycle:

$$E_{\text{rec}} = \int_{t_0}^{t_1} \eta_H [P_t - P_c - J\omega\dot{\omega} - P_{\text{loss}}]_+ dt. \quad (8)$$

Here η_H combines machine and inverter efficiency. Motor operation must be subtracted separately to obtain net electrical energy. Equations (7)–(8) show why maximizing a single steady-state power value is insufficient: electrical extraction can reduce turbocharger acceleration and therefore future engine power.

4 Prototype and experimental method

4.1 Mechanical design

The turbine wheel and volute were modelled in SolidWorks. SolidWorks Flow Simulation and CFTurbo were used as preliminary design tools. The archived files do not document a mesh-independence study, calibrated boundary conditions or a measured turbine map, so the simulations are retained as qualitative design evidence rather than quantitative validation.

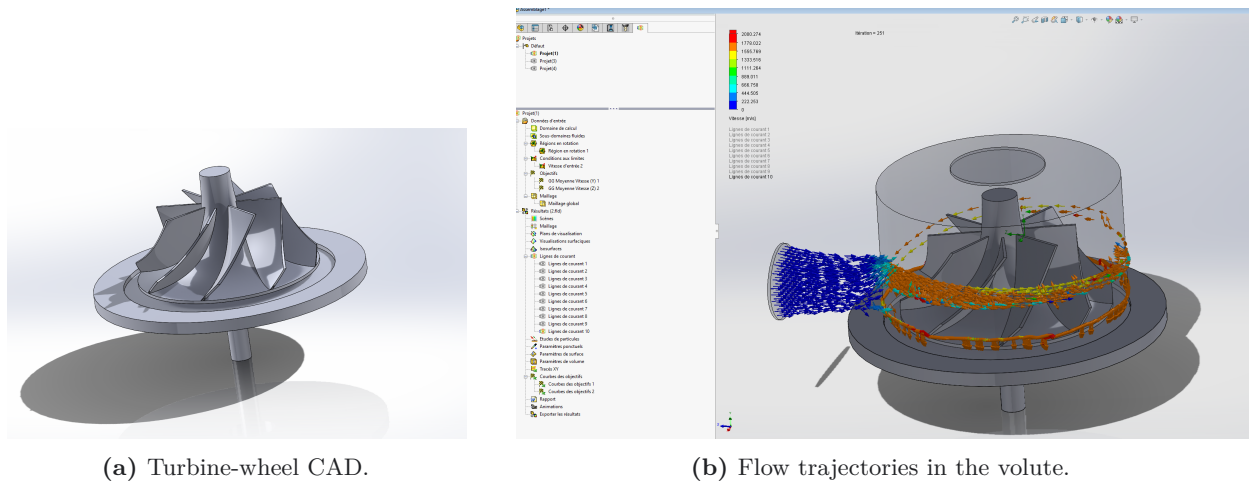


Figure 7. Mechanical design and preliminary flow visualization.

Two printed prototypes were produced. The first used a small machine and a separate printed shaft. Its failure exposed the importance of bending stiffness, balance and stress concentration. The final version mounted the wheel directly on the shaft of a larger permanent-magnet DC machine.

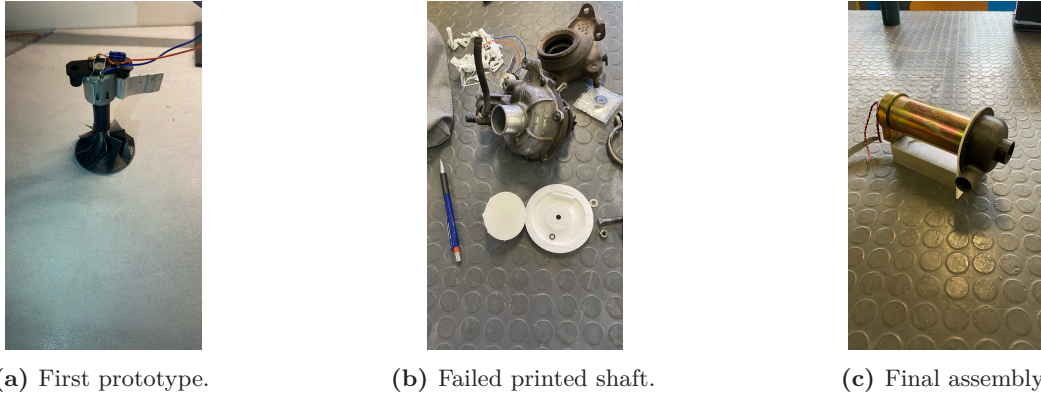


Figure 8. Development of the turbine-generator demonstrator.

4.2 Experimental arrangement

Figure 9 is a clean reconstruction of the mounting diagram shown on slide 20. Compressed air provides the aerodynamic input. The DC machine supplies a resistor R . An oscilloscope measures load voltage with a 10:1 probe, while a shaft encoder sends position increments to a Teensy microcontroller and then to the acquisition computer.

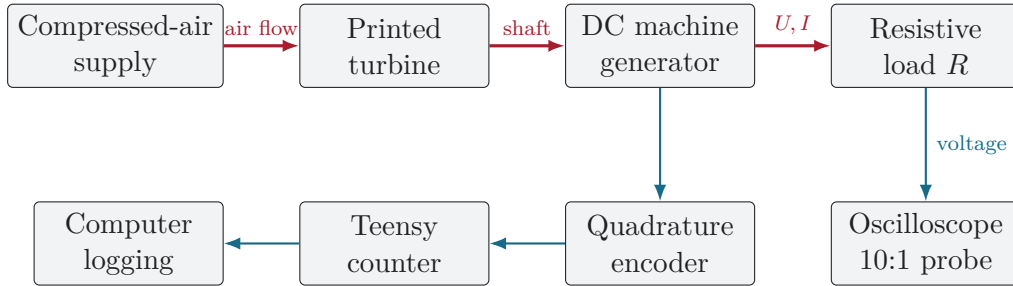


Figure 9. Experimental setup and measured signals.

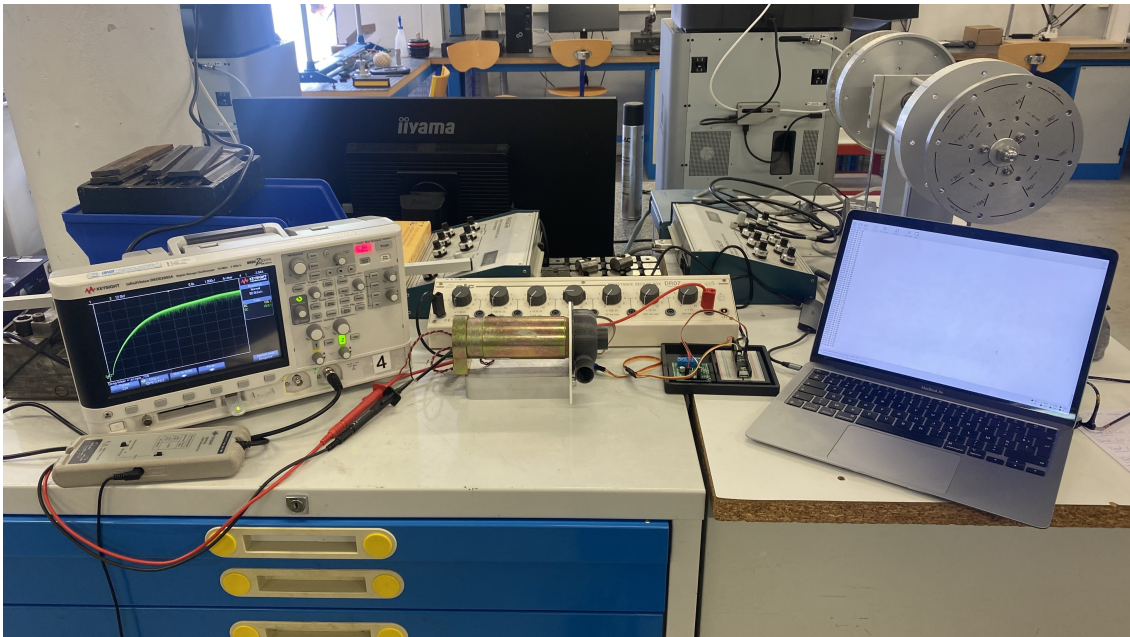


Figure 10. Experimental bench used for the resistance sweep.

The air supply was kept at the same nominal setting while R was changed from $100\ \Omega$ to $3\ \text{k}\Omega$.

Voltage and speed were allowed to approach a plateau. The Arduino program evaluates encoder counts every 10 ms; its archived output is in degrees per second and is converted to rpm by division by six.

4.3 Electrical machine model

For the permanent-magnet DC machine,

$$e = k_e \omega, \quad C_e = k_t i, \quad i = \frac{e}{R + r}. \quad (9)$$

The steady mechanical balance under approximately constant drive torque C and viscous friction $\mu\omega$ is

$$C = \mu\omega + k_t \frac{k_e \omega}{R + r}. \quad (10)$$

Taking $k_e = k_t = k$ in SI units gives

$$e = \frac{C}{\mu/k + k/(R + r)}. \quad (11)$$

The useful voltage and power are

$$U = \frac{R}{R + r} e, \quad P_R = \frac{U^2}{R} = \frac{e^2 R}{(R + r)^2}. \quad (12)$$

Equation (12) is preferable to e^2/R unless e explicitly denotes load voltage or $r \ll R$.

Defining $q = \mu/k^2$ and $B = C/k$ gives a useful loss decomposition:

$$P_R = \frac{B^2 R}{[1 + q(R + r)]^2}, \quad P_J = \frac{B^2 r}{[1 + q(R + r)]^2}, \quad (13)$$

$$P_f = \frac{q B^2 (R + r)^2}{[1 + q(R + r)]^2}. \quad (14)$$

Here $P_J = r i^2$ and $P_f = \mu \omega^2$. Differentiating P_R yields

$$R_\star = r + \frac{1}{q} = r + \frac{k^2}{\mu}. \quad (15)$$

The optimum is therefore not the electrical maximum-power condition $R = r$ of an ideal fixed-speed source. Rotor speed changes with load, and mechanical friction participates in the optimum.

5 Results

5.1 Measured operating points

The presentation table reported power values from 190 to 600 with a watt label. Those values cannot be watts: the selected machine is rated at approximately 17–19 W. Interpreting them as milliwatts gives voltages consistent with the 10:1 probe. For example,

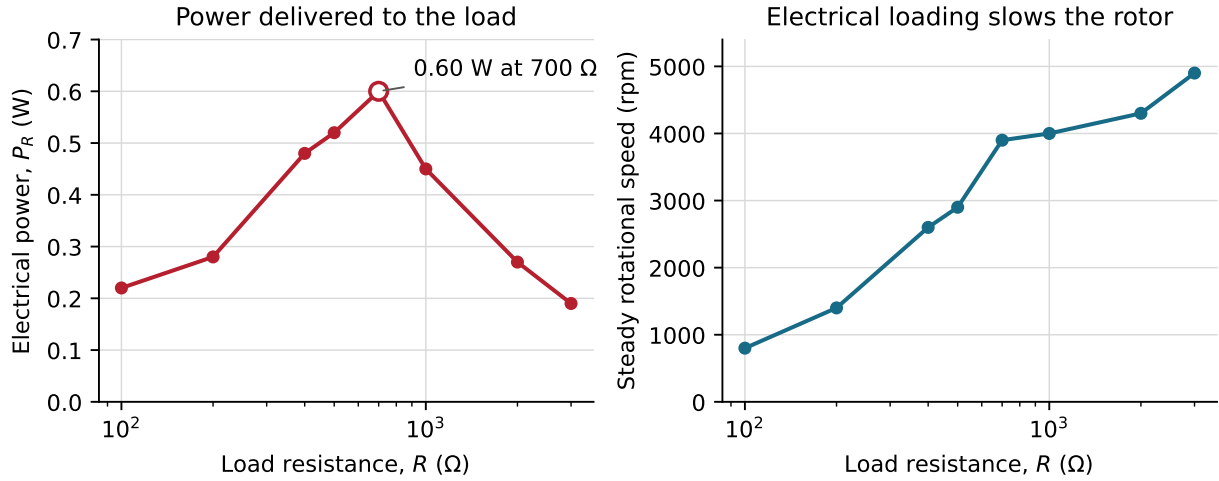
$$U = \sqrt{P_R R} = \sqrt{0.600 \text{ W} \times 700 \Omega} = 20.5 \text{ V}.$$

The report therefore preserves the numerical measurements and corrects the unit to watts.

Table 1. Steady values from the resistance sweep.

R (Ω)	Speed (rpm)	U (V)	I (mA)	P_R (W)
100	800	4.69	46.9	0.220
200	1400	7.48	37.4	0.280
400	2600	13.86	34.6	0.480
500	2900	16.12	32.2	0.520
700	3900	20.49	29.3	0.600
1000	4000	21.21	21.2	0.450
2000	4300	23.24	11.6	0.270
3000	4900	23.87	8.0	0.190

Rotor speed rises monotonically with resistance, from approximately 800 rpm at $100\ \Omega$ to 4900 rpm at $3\text{ k}\Omega$. This is direct evidence that generator current produces a significant braking torque. Recovered power is non-monotonic and reaches 0.60 W at $700\ \Omega$.

**Figure 11.** Measured electrical power and steady rotor speed.

5.2 Interpretation of Joule and friction losses

Figure 12 reconstructs and extends the analysis shown on slide 21. The analytical curve uses the measured motor armature resistance $r = 6.98\ \Omega$. The ratio $q = \mu/k^2$ is selected so that Equation (15) reproduces the original prediction $R_* \simeq 800\ \Omega$, and the scale B is chosen to give the slide's predicted peak of 0.623 W . It is therefore an interpretation of the original model, not an independent fit of all measured points.

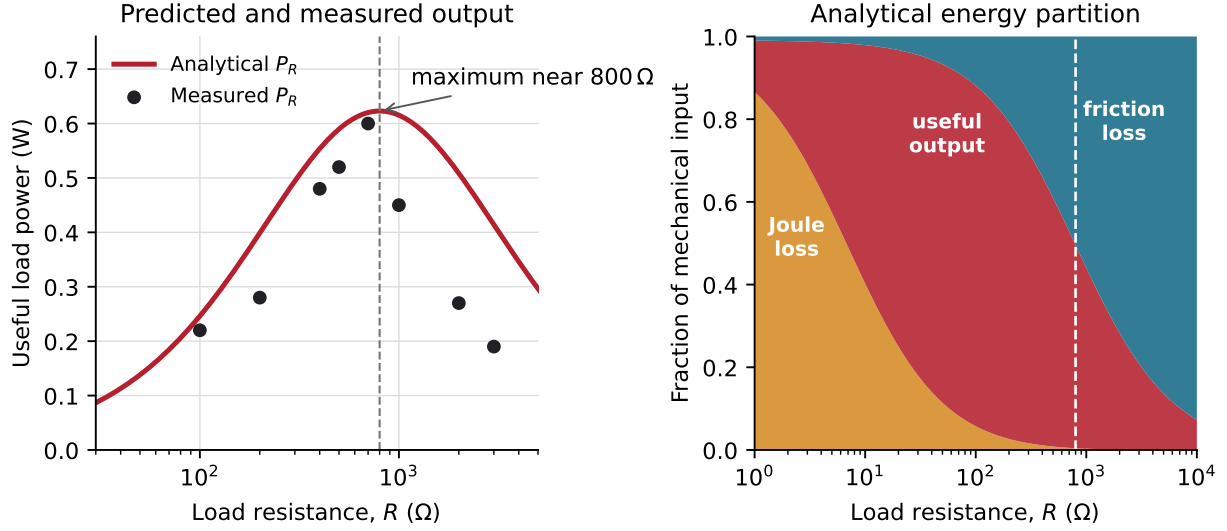


Figure 12. Useful power and analytical loss partition from the slide 21 model.

Three regimes explain the curve:

- At low R , current is high. Internal copper loss $P_J = ri^2$ increases, and electromagnetic torque ki slows the rotor. The lower speed reduces $e = k\omega$ and therefore suppresses useful power.
- Near 700–800 Ω , current remains sufficient while the rotor has recovered enough speed to generate about 20 V. Useful power is maximal.
- At high R , the machine approaches open circuit. The rotor spins faster, but current and $P_R = U^2/R$ fall. Under the constant-torque model, an increasing share of the input is dissipated by bearing, windage and other speed-dependent mechanical losses.

The predicted maximum of 0.623 W near 800 Ω differs by about 3.7 % in magnitude from the measured maximum of 0.600 W at 700 Ω . The agreement is encouraging, but the load grid is sparse and the friction parameter was not identified independently. The defensible experimental conclusion is an optimum in the 500–1000 Ω interval for this air-supply condition.

5.3 Uncertainty

No calibrated uncertainty budget or repeated-run statistics were stored. If voltage and resistance uncertainties are independent, first-order propagation gives

$$\left(\frac{\sigma_P}{P}\right)^2 = \left(2\frac{\sigma_U}{U}\right)^2 + \left(\frac{\sigma_R}{R}\right)^2. \quad (16)$$

Numerical confidence intervals would additionally require oscilloscope accuracy, resistor tolerances, encoder uncertainty, air-flow repeatability and repeated tests. They are not invented here.

6 Discussion

6.1 What the demonstrator establishes

The experiment confirms the central coupling of the MGU-H problem: electrical extraction changes shaft speed. A nearly open circuit preserves speed but produces little power. A strong electrical load draws current but can collapse speed and voltage. The controlled inverter of a real MGU-H acts as a variable electromagnetic load and moves the operating point as exhaust flow and boost demand change.

In a Formula One controller, the optimum is subject to compressor surge and choke limits, turbo speed, thermal limits, energy-store state of charge, requested crankshaft torque and the value of

energy later in the lap. The objective is not simply maximum instantaneous P_H ; it is minimum lap time or minimum fuel use under constraints. A generator command that delays boost can cost more propulsion energy than it recovers.

6.2 Limits of the scale model

The demonstrator speed is more than an order of magnitude below the 125 000 rpm limit specified for the original MGU-H [3]. Compressed air replaces pulsating hot exhaust gas. Turbine input torque, mass flow and pressure ratio were not measured; mechanical input power and total efficiency therefore cannot be calculated. The printed geometry has no calibrated map, and only one point was retained for each resistor.

A quantitative continuation would add a regulated pressure supply, mass-flow and pressure sensors, simultaneous voltage-current acquisition, repeated runs, coast-down identification of $P_f(\omega)$ and an electronic load. The final step would replace resistance control by closed-loop torque control and optimize Equation (8) on measured engine and turbocharger maps.

6.3 From bench resistance to F1 energy management

The bench load R is an analogue of inverter torque demand. Increasing generator current recovers more power only until the induced braking moves the shaft away from its useful operating point. The same principle connects the MGU-H to the rest of the Formula One electrical system:

- during a throttle transient, stored energy is spent through the MGU-H to protect compressor response;
- at high exhaust power, MGU-H torque replaces wastegate dissipation and supplies the DC link;
- when propulsion is valuable, this power can feed the MGU-K directly;
- when later deployment is more valuable, it can charge the energy store, subject to efficiency and state-of-charge limits.

Published vehicle studies similarly find that a shaft-integrated motor-generator can combine low-speed assistance with high-load recovery, but the net benefit depends on component efficiency and drive cycle [7–9].

7 Regulatory perspective

Formula One removed the MGU-H from the 2026 power-unit architecture. The new rules retain the turbocharged V6 and increase MGU-K electrical power from 120 kW to 350 kW, shifting the hybrid balance toward kinetic recovery and battery deployment [10]. The removal reflects cost, complexity and road-relevance objectives; it does not invalidate the thermodynamic principle. Electrically assisted turbochargers continue to transfer the same idea to road vehicles, where packaging, durability, cooling and system voltage determine viability.

8 Conclusion

This work places the experimental turbine-generator bench within the complete electrical architecture of the 2014–2025 Formula One power unit. The MGU-K recovered braking energy and returned torque to the crankshaft. The MGU-H acted on the turbocharger shaft: motor operation reduced turbo lag when turbine power was insufficient, and generator operation recovered excess shaft power when turbine output exceeded compressor demand.

The original thermodynamic curve makes this transition visible. The shaft balance shows that recoverable power is the turbine-compressor surplus after rotor acceleration and mechanical losses. The resistance sweep then demonstrates the associated electromechanical trade-off. The best sampled point is 700 Ω , 3900 rpm and approximately 0.60 W; the analytical model predicts a broad maximum near 800 Ω . Low resistance raises current, Joule loss and braking torque. High resistance reduces current while mechanical losses absorb an increasing share of the available input.

The principal conclusion is consequently a control result: useful recovery is maximized by coordinating electrical torque with turbocharger speed and future boost demand, not by maximizing generator current at every instant. In the Formula One system, that command must also be coordinated with the MGU-K, power electronics and energy store over the complete lap.

Data and code availability

The project archive contains the original presentation, Python models, Arduino acquisition code, oscilloscope CSV files, encoder records, CAD results and calculation workbooks. The report source and publication-figure script are stored in the `report` directory.

A Parameters used in the reconstructed thermodynamic calculation

Table 2. Illustrative mean-value model parameters.

Parameter	Symbol	Value
Engine displacement	V_d	1.6 L
Air density	ρ_a	1.2 kg m^{-3}
Volumetric efficiency	η_v	0.90
Air–fuel ratio	AFR	14.7
Compressor inlet temperature	T_1	300 K
Turbine inlet temperature	T_3	$800 + 0.05(N - 1000) \text{ K}$
Pressure ratio	$p_2/p_1 = p_3/p_4$	3.0
Compressor efficiency	η_c	0.72
Turbine efficiency	η_t	0.70

B Reproducibility notes

The archived Arduino program uses a nominal 10 ms sampling period and 4000 encoder increments per revolution. Some raw files contain isolated zeros and spikes near start-up and shutdown. Table 1 reproduces the consolidated values from the presentation rather than applying a new undocumented selection rule. The resistance-to-filename mapping in the encoder archive is not explicit, so no new quantitative transient comparison is claimed.

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